

BHAVISH POTHURAJU

+91 7337421621 [◇ Github](#) [◇ LinkedIn](#) [◇ bhavishpothuraju123@gmail.com](#) [◇ Bengaluru, India](#)

EDUCATION

Bachelor of Technology in Computer and Communication Engineering

Sep 2020 – Jun 2024

Manipal University Jaipur, Jaipur

CGPA: 8.15/10

Minor Specialization: Computational Intelligence and Soft Computing

TECHNICAL SKILLS

- **Programming Languages & LLM Frameworks:** Python - NumPy, Pandas, Matplotlib, C++, SQL, PyTorch, Hugging Face Transformers, LangChain, LlamaIndex, RAG Pipelines, LoRA / QLoRA, PEFT
- **Cloud & MLOps Tools:** AWS SageMaker, S3, EC2, Lambda, Bedrock, Docker, GitHub Actions, Git / GitHub
- **Coursework:** Machine Learning, Deep Learning, Computer Vision, Digital Image Processing, Natural Language Processing, Transformers, Generative AI, Data Structures & Algorithms, Statistics, Relational databases, OOP

EXPERIENCE

Data Scientist-1

Dec 2024 - Present

SigTuple

Bengaluru

- Fine-tuned **LLaMA-2** large language model using **LoRA** and **QLoRA** on domain-specific clinical hematology data, using **Hugging Face** libraries with **PEFT** and containerized the pipeline using **Docker**, pushing images to **Amazon ECR** and deploying the application on an **AWS EC2**.
- Designed and deployed **YOLOX-Nano** and **YOLOX-Small** multi-object detection models for RBCs, WBCs, and Platelets, improving test precision from **95% to 99.9%** and recall to **97%**.
- Implemented a **false coloring pipeline** by **superimposing UV and Brightfield microscopic images** using OpenCV, enhancing cell visibility, using **multi-space color conversions**.
- Integrated **NATS JetStream pub-sub** to connect image reconstruction and extraction microservice, allowing message exchange with the main orchestrator in the pipeline.
- Used **Segment Anything Model (SAM)** to generate initial cell segmentation masks, which were refined and used to train a lightweight **U-Net** model, **achieved 94%** segmentation accuracy and reducing the computation load and time by **20%**.
- Built and optimized **ResNet-based** classification models for Quality Checking FOVs explored multiple training strategies (**full fine-tuning and head-only, base freezing**), achieving up to **90% precision and recall**.

Data Science Intern

June 2024 - Nov 2024

SigTuple

Bengaluru

- Extracted hematological metrics (MCV, RDW, MCH) from microscopy images using **Image Processing techniques**, **Otsu/Adaptive thresholding**, and **Gaussian blur + Canny edge detection**, with **$R^2 = 0.94$** .
- Visualized feature embeddings via **3D feature vector plots**, **histogram analysis**, and **regression line fitting**, enabling better model interpretability and clinical parameter correlation.
- Conducted **CPU performance profiling** and **parallelized model inference pipelines** using **Python Process and Thread Pool Executor**, leading to sustained **98–100% CPU utilization** across multi-core systems.
- Applied **core affinity optimization** to restrict computational workloads to specific cores, reducing system-level contention and improving throughput of concurrent image processing tasks.

PROJECTS

Conversational RAG App for PDF QA [Link](#)

Mar 2025 – Apr 2025

Built a full-stack Retrieval-Augmented Generation (RAG) system using **LangChain**, **Groq API (Gemma-2b)**, **ChromaDB**, and **Streamlit** to answer PDF-based queries. Integrated **context summarization** and vector store retrieval using **HuggingFace embeddings** with **Chat history** and containerized the application.